INCH POUND
MIL-DTL-43549D
18 December 2000
SUPERSEDING
MIL-C-43549C
23 June 1989

DETAIL SPECIFICATION

CAP, COLD WEATHER, INSULATING, HELMET LINER

This specification is approved for use by all Departments and Agencies of the Department of Defense in preference to MIL-C-43549.

1. SCOPE

1.1 Scope. This specification covers a fully lined cap cold weather, insulating helmet liner, with adjustable forehead flap.

1.2 Classification. The woodland camouflage pattern cap will be one Type in the following sizes as specified (see 6.2).

Schedule of sizes

6-1/2 6-3/4 7 7-1/4 7-1/2 7-3/4 7-7/8 8

2 APPLICABLE DOCUMENTS

SPECIFICATIONS

FEDERAL

A-A-50198	- Thread, Gimp, Cotton, Buttonhole
A-A-55126	- Fastener Tapes, Hook and Loop, Synthetic

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center Philadelphia, Clothing and Textiles Directorate, ATTN: DSCP-CRFD, 700 Robbins Avenue, Bldg 6D, Philadelphia, PA, 19111-5096, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or letter.

AMSC N/A FSC 8415

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

A-A-50199	- Thread, Polyester Core: Cotton-, Rayon-, or Polyester-Covered
A-A-52074	- Cloth, Plain Weave, Acrylic
JJ-W-155	- Webbing, Textile, (Cotton, Elastic)
A-A-55234	- Cloth, Knitted, Nylon, Fleece

MILITARY

MIL-T-3530	- Thread and Twine; Mildew Resistant or Water Repellent Treated
MIL-C-3924	- Cloth, Oxford, Cotton Warp and Nylon Filling, Quarpel Treated
MIL-W-5664	- Webbing, Textile, Elastic
MIL-C-43247	- Cloth, Knitted, Nylon, Tubular, Stretch Type
MIL-DTL-32075	- Label: For Clothing, Equipage, and Tentage, (General Use)

2.1 Government publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Federal Regulations:

Federal Trade Commission

Rules and Regulations Under the Fur Products Labeling Act

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

Rules and Regulations Under the Textile Fiber Products Identification Act

Rules and Regulations Under the Wool Products Labeling Act of 1939

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless specific issue is identified, the issue in effect on date of invitation for bid or request for proposal will apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-6193 - Standard Practice for Stitches and Seams

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

COLOR ASSOCIATION OF THE UNITED STATES

Standard Color Card of America

(Standard shades may be available from the Color Association of the United States, Inc., 589 8th Avenue, New York, NY 10036. If standard shades are not available from the Color Association, individual color samples may be obtained from the contracting activity or as directed by the contracting activity.)

American Society For Quality Control

ANZI/ASQC Z1.4 – Sampling Procedures and Tables for Inspection of Attributes

(Applications for copies should be addressed to the American National Standards Institute, 11 42nd Street, New York, NY 10036.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Recycled, recovered, or environmentally preferable material. Recycled, recovered, or environmentally preferable materials are used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.1 Basic material. The basic material for the outer shell (crown, earlap, forehead flap, and binding) will be cotton warp and nylon filling oxford cloth, conforming to class 3 of MIL-C-3924. The cloth for the sweatband will be either class 3 or class 2 dyed Camouflage Green 483 material of MIL-C-3924. As an alternative, the cloth may be printed seconds, or dyed the ground

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

shade prior to printing of class 3 material of MIL-C-3924, or dyed seconds of class 2 material of MIL-C-3924. Printing seconds will be cloth which has been rejected only for defects cited in MIL-C-3924 pertaining to shade, infrared reflectance, or pattern of camouflage print. Dyed seconds shall be cloth which has been rejected only for defects cited in MIL-C-3924 pertaining to shade, and the shade will approximate Camouflage Green 483.

3.2.2 Lining for crown. The material for lining the crown will be cloth satin lining conforming to A-A-52074, and the shade shall approximate Green 483.

3.2.3 Lining for earlap and forehead flap. The material for lining the earlap and forehead flap will be knitted nylon fleece cloth conforming to Type I of A-A-55234 except that the color shall be Camouflage Green 483. There will be a minimum of 24 wales per inch, and the minimum colorfastness to laundering (after 3 cycles) shall be "fair."

3.2.4 Gusset material. The material for the gusset will be nylon knitted stretch type cloth, Camouflage Green 483, conforming to class 2 of MIL-C-43247.

3.2.5 Back adjustment strap. The strap for outside back of earlap will be cotton elastic webbing, 1 inch wide, Camouflage Green 483, conforming to Type I, class 7 of JJ-W-155.

3.2.6 Earlap fasteners. The material for the adjustable closure system will be nylon fastener tape, 1-1/2 inch wide loop tape and 1-1/2 inch hook tape, conforming to Type II, class 1 of A-A-55126. The color will be Camouflage Green 483.

3.3. Thread, polyester core. The thread for stitching will be cotton- covered or polyester-covered polyester thread conforming to A-A-50199. The thread size will be Ticket Nos. 50-2 and 70-2. The thread will be dyed S-1, C.A. 66022 and water repellent treated in accordance with Type II, classes 2 and 3 respectively of MIL-T-3530. Lubrication of the quarpel treated thread by any means in the end item fabricators plant prior to or during sewing is prohibited.

3.3.1 Gimp. The gimp for reinforcing the buttonholes shall conform to Type I or II, size No. 8, conforming to A-A-50198. The gimp shall be dyed S-1, C.A. 66022.

3.4 Labels. Each cap will have a combination identification-size label and an instruction label conforming to MIL-DTL-32075. Label color will be Medium Green Cable No. 70034, 70130, or 70131 and will show fastness to laundering in accordance with MIL-DTL-32075 requirements.

3.4.1 Identification-size label. The identification-size label will conform to Type VI, class 4.

3.4.2 Instruction label. The instruction label will conform to Type VI, class 3, and will contain the following information:

CAP, COLD WEATHER, INSULATING, HELMET LINER

1. Wear straight on head with forehead flap up or down.
2. Fasten earlaps under or over chin, at back of neck, or over head.
3. When worn under helmet with earlaps over helmet, thread helmet chin strap through buttonholes.
4. Launder by the Wool Method, Formula II, FM 10-16 except dry tumble heat shall not exceed 130°F; or hand wash in lukewarm water.

DO NOT REMOVE THIS LABEL

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

3.4.3 Label/tag. Each item will be individually bar coded with a paper tag for personal clothing items. The paper tag will be standard bleached sulfate having a basis weight of 100 pounds. The paper used for the tags will have a smooth finish to accept thermal transfer and direct printing. The tags will have a hole and will be attached to each item by a fastener, clearly legible and readable by scanner. The bar coding element will be a 13 digit national stock number (NSN). There will be a 12 digit unit price control number (UPC) assigned for NSN's by the procuring agency. The initial letters UPC must appear beneath the code. The bar code NSN and UPC type will be a medium to high code density and will be located so that it is completely visible on the item when it is folded an/or packaged as specified and so that it causes no damage to the item. The UPC code must also be placed on all shipping cartons on which the NSN appears.

3.5 Design. The cap is a fully lined cold weather head covering having a turndown Forehead flap (sinus protector), three-piece crown, a stretch gusset with an elastic strap at center back and earlaps with nylon fastener hook and pile tape overlap closure (see Figure 1).

3.6 Patterns. Standard patterns, which provide a 3/16 inch seam allowance for joining forehead flap to flap lining and 3/8 inch for all other seams, will be furnished by the Government. The Government patterns, which show directional lines and markings for proper assembly, will not be altered in any way and are to be used as a guide for cutting contractor's working patterns.

The working patterns will be identical to the Government patterns except that additional notching to facilitate manufacture is permissible. Also, minor modifications are permitted where necessary when using automatic equipment. These modifications shall not alter the dimensional, serviceability, or appearance requirements.

3.6.1 Pattern parts. The pattern parts and applicable material for cutting the cap parts will be as specified in Table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

TABLE I. List of pattern parts

Material	Nomenclature	Cut parts
Cotton warp and nylon filling oxford cloth ‘	Top crown	1
	Side crown	2
	Earlap	2
	Forehead flap	1
	Sweatband 1/	1
Satin cloth	Top crown lining	1
	Side crown lining	2
Nylon fleece knitted cloth	Earlap lining	2
	Forehead flap lining	1
Nylon stretch type knitted cloth	Gusset	2
Nylon fastener tape	Loop tape (Inside left earlap - (upper portion)	1
	Loop tape (inside left earlap - (lower portion)	1
	Hook tape (outside right earlap - (upper portion)	1
	Hook tape (outside right earlap - (lower portion)	1

1/ When the sweatband material is strip cut and put on rolls, disregard this pattern piece. in Table II and herein. Figure 1 is furnished solely for guidance and information. Should variation from the document appear in Figure 1, the document will govern.

3.7 Stitches, seams, and stitching. All stitches, seams, and stitching will conform to ASTM-D-6193. The type of seam, stitching, and stitches per inch will be as specified in Table II. Seam allowances shall be maintained with seams sewn, so that no raw edges, run-offs, pleats, puckers, or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used. Where stitch type 401 is used, the chain or underside of the stitch shall be on the inside of the cap.

3.8.1 Type 301 stitching. Ends of all stitching will be backstitched or overstitched not less than 1/4 inch, except where ends are turned under or caught in other seams or stitching. Ends of a continuous line of stitching shall overlap not less than 1/2 inch. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock will be embedded in the materials sewn.

3.8.2 Repairs of type 301 stitching. Repairs of type 301 stitching will be as follows:

a. When thread breaks, skipped stitches, run-offs, or bobbin runouts occur during sewing, except prestitching, the stitching will be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

b. Except for prestitching, thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item will be repaired by overstitching. The stitching will start a minimum of 1/2 inch in back of the defective area, continue over the defective area, and continue a minimum of 1/2 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching will be repaired by removing the defective stitching without damaging the materials, and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.8.3 Type 401, 504, and 505 stitching. Thread tension will be maintained so that there will be no loose or excessively tight stitching. All repairs shall be in accordance with 3.8.2. a and b. Type 301 stitching may be used to repair Type 401 stitching.

3.8.4 Buttonhole. The buttonholes will be eyelet end, taper bar type, worked over gimp (see 3.3.1) with not less than four tacking stitches at bar end (not counting crossover stitch). The purling shall be on the outside. The finished cut lengths shall be 1 to 1-1/8 inches. The buttonholes shall be clean cut with the stitching securely caught 'in the fabric. Additive treatments or materials used to secure the ends of the stitching or to stiffen the fabric, prior to stitching the buttonholes are prohibited.

3.8.5 Bartacking. Bartacking will be 1/2 $\pm$ 1/16 inch long, 1/8 $\pm$ 1/32 inch wide and shall contain a minimum of 28 stitches. Bartacks shall be free from thread breaks and 1oose stitching.

3.8.6 Automatic stitching. Automatic stitching machines may be used to perform any of the required stitch patterns provided the requirements of the stitch pattern, stitches per inch, and size and type of thread are met, and at least three tying, overlapping, or back stitches are used to secure the ends of the stitching.

3.8.7 Thread ends. All thread ends that are visible on the finished item will be trimmed to a length of not more than 1/4 inch.

3.8.8 Repairs. Repairs such as mends, darns, patches, or splices are not permitted on the cap.

3.9 Manufacturing operations requirements. The cap will be manufactured in accordance with operations requirements specified in Table II. Unless otherwise specified in Table II, the contractor is not required to follow the exact sequence of operations listed. Any additional basting or holding stitching to facilitate manufacture is permissible provided the thread is removed or does not show on the finished cap.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

NO.	TABLE II MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STICHING TYPE	STITCHES PER INCH	NEEDLE	THREAD BOBBIN/ LOOPER/
1.	Cutting. a. Cut the material in strict accordance with patterns furnished which show size, shape, warp or filling directional lines, and notches for proper assembly. The side crown lining pieces may be cut in either the warp or filling direction. The napped side of the crown lining pieces shall face the outer shell in the finished cap. b. Cut the basic material parts from the same piece of material except the forehead flap and binding, which may be cut from ends. Cut the lining for forehead flap and earlaps from the same piece of material. c. Cut the binding for earlap from the basic material, on the bias and sufficiently wide to finish as specified in operation 9. The sweatband material may be strip cut (on the bias) to the proper width and placed on rolls. It may be pieced with no more than one joining seam in the finished cap. When applicable, the joining seam shall be stitched 3/16 to ¼ inch, using stitch type 301 or 401 and seam type SSa-1. d. Cut the gusset pieces so that wales run the length of the gusset (vertically). (The wales of the knitted fabric are equivalent to warp yarns of the woven fabric.) e. Cut the forehead lining (fleece) in either the warp or filling direction.					

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

N0.	TABLE II (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STICHING TYPE	STITCHES PER INCH	NEEDLE	THREAD BOBBIN/ LOOPER/
1.	Cutting. (cont'd) f. Cut the 1-inch wide elastic webbing for the back adjustment strap (one each) of sufficient length to finish 2 to 2-1/4 inches when attached to cap (see operation 14). g. Cut the nylon fastener tapes by patterns.					
2.	Replacement of damaged parts. During the spreading, cutting, and manufacturing process, components having defects or damages that are classified as defects in 4.4.3 shall be removed from production and replaced with non-defective and properly matched components.					
3.	Component marking. a. Any method of marking may be used except: 1. Corrosive metal fastener devices 2. Sew-on tickets 3. Adhesive tickets which leave an impression or traces of paper or adhesive on the material after removal of the ticket.					
4.	Make forehead flap. Finished appearance. The forehead flap shall be smooth and free of wrinkles and puckers. Raised stitching shall be neat and evenly gaged.					

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

N0.	TABLE II (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STICHING TYPE	STITCHES PER INCH	NEEDLE	THREAD BOBBIN/ LOOPER/
4.	Make forehead flap. (cont'd)					
	a. Superimpose the forehead flap and flap lining, face to face, and stitch the plies together along the curved edge; force out and raise stitch 1/8 to 14 inch from edge.	301, or 401 and 301	SSe-2 (a) SSe-2 (a) SSe-2 (b)	10-14 10-14 10-14	50-2 50-2 50-2	50-2 70-2 50-2
	b Overedge the bottom raw edges together. The gage shall be 3/16 to 1/4 inch wide, and a 1/2 to 1 inch chain shall be left at each end.	504 or 505	SSa-1 SSa-1	10-14 10-14	70-2 70-2	70-2 70-2
5.	Make outer shell. Finished appearance. The outer shell shall be neatly joined without twists, puckers, or pleats and with joining seams matching. Darts shall be equal in length ± 1/8 inch on each side, and crown shall be shaped. Raised stitching shall be neat and evenly gaged.					
	a. Stitch the out darts of the side crown pieces and earlaps, tapering to nothing 1/2 to 5/8 inch beyond the cutout.	301	SSa-1	10-14	50-2	50-2
	b. Stitch the folded darts on the earlaps tapering to nothing 1/2 to 5/8 inch beyond end of dart.	301	OSf-1	10-14	50-2	50-2
	c. Seam side crowns to earlaps leaving the seam open at the front ends 1/2 to1 inch to allow for binding.	301 or 401	SSa-1 SSa-1	10-14 10-14	50-2 50-2	50-2 70-2
	d. Stitch top crown to gusset; turn seam toward crown and raise stitch 1/2 to 3/16 inch from folded edge.	301or 401 and 301	LSq-2 (a) LSq-2 (a) LSq-2 (b)	10-14 10-14 10-14	50-2 50-2 50-2	50-2 70-2 50-2
	e. Stitch top crown, with gusset attached, to side crowns and back edges of earlaps.	301, or 401	LSq-2 (a) LSq-2 (a)	10-14 10-14	50-2 50-2	50-2 50-2

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

N0.	TABLE II (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STICHING TYPE	STITCHES PER INCH	NEEDLE	THREAD BOBBIN/ LOOPER/
5.	Make outer shell. (cont'd)					
	f. Slash seam at top of gusset; turn crown seam toward center; turn gusset seams toward front of crown. Raise stitch around crown 1/8 to 3/16 inch from folded edge beginning at face opening, extending to and across gusset seam, ending at opposite crown seam face opening.	301	LSq-2(b)	10-14	50-2	50-2
	-or –					
	g. Raise stitch crown 1/8 to 3/16 inch from folded edges, the raise stitch gusset seam 1/8 to 3/16 inch from folded edge.	301	LSq-2(b)	10-14	50-2	50-2
6.	Make perforations in earlap linings. Make perforations in each earlap lining as indicated by marks on pattern. The perforations shall be free of ridges and rough or hard edges.					
7.	Make lining and attach labels. Finished appearance. The lining pieces shall be neatly joined without twists, puckers, or pleats and with joining seams matching the outer crown pieces. The top and side pieces shall be well shaped. The labels shall be centered, straight, and the stitching shall not be through the printing.					
	a. Stitch folded darts of the side crown tapering to nothing 1/2 to 5/8 inch beyond end of dart.	301	OSf-1	10-14	50-2	50-2
	b. Center instruction label on right side Of crown lining and the combination identification-size label on the left side crown lining and stitch on all four sides 1/8 ± 1/16 inch from edge. The labels shall be on the unnapped surface. The stitching shall not go through the printing.	301	LSbj-1	8-14	50-2	50-2

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

N0.	TABLE II (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STICHING TYPE	STITCHES PER INCH	NEEDLE	THREAD BOBBN/ LOOPER/
7.	Make lining and attach labels. cont'd)					
	c. Stitch gusset lining piece to top crown lining piece.	301 or 401	LSq-2 (a) LSq-2 (a)	10-14 10-14	50-2 50-2	50-2 70-2
	d. Precrease sweatband 3/8 ± 1/6 inch along top edge and 5/8 ± 1/16 inch along bottom edge. The sweatband shall measure 1 ± 1/18 inch wide on the finished cap.					
	e. Stitch top crown lining and attach gusset to side crown lining pieces and earlap linings with each end of the sweatband caught in the seaming. The bottom edge of the side crown lining pieces shall be inserted between the 5/8 inch fold of the sweatband. The earlap linings shall overlap the bottom edge of sweatband 3/8 inch. The ends of the sweatband and earlaps may be tacked to hold in position. The side crown darts shall be turned toward the center crown.	301or 401	LSq-2 (a) LSq-2 (a)	10-14 10-14	50-2 50-2	50-2 70-2
	f. Turn seams toward top crown and raise stitch 1/8 to 3/16 inch across back both sides of top crown. The raised stitching may extend to bottom of sweatband.	301	LSq-2(b)	10-14	50-2	50-2
8	Attach sweatband.					
	a. Stitch the top folded edge of sweatband to the crown lining 1/16 to 1/8 inch from folded edge.	301	LSd-1	10-14	50-2	50-2
	b. Insert the bottom edge of crown lining between the 5/8 inch fold of sweatband and stitch to crown lining 1/16 to 1/8 inch from raw edge of sweatband. NOTE: The sweatband need not be precreased (operation 7.d) if conformance to all other requirements has been met.	301	SSw-2(a)	10-14	50-2	50-2

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

NO.	TABLE II (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STICHING TYPE	STITCHES PER INCH	NEEDLE	THREAD BOBBIN/ LOOPER/
9.	Attach earlap lining to earlap shell. Finish appearance. The earlap shall finish without pleats or puckers. Binding shall be neat and uniform in width with stitching evenly gaged.					
	a. Place earlap lining on the inside of earlap shell and stitch raw edges together 1/8 to 3/16 inch from the edge along sides and across bottom. NOTE: When an overedge stitch is used, the guides and knives shall be set to trim only raveled ends of the fabric.	301, 401 504, or 505	SSa-1 SSa-1	10-14 10-14	50-2 50-2	50-2 70-2
	b. Turn in raw edges of bias bindings (see operation 1.c) and bind sides and bottom	301	BSc-1	10-14	50-2	50-2
	- or -					
	c. Stitch one side of the binding to sides and bottom edges on the inside of cap.	301	BSg-2(a)	10-14	50-2	50-2
	- and -					
	d. Fold binding to outside, turn in raw edges and stitch 1/16 to 3/32 inch from edge. NOTE: The binding shall finish 5/16 ± 1/16 inch wide.	301	BSg-2(b)	10-14	50-2	50-2
10	Join top of earlap lining to earlap shell. Seam earlap to crown at front ends catching the binding securely in stitching (see operation 5.c.). Continue the stitching, seaming the earlap lining to the earlap crown joining seam, with the row of stitching superimposed or adjacent to stitching of ear lap-to-crown joining seam (see operation 5.c.).	301 or 401	SSa-1 SSa-1	10-14 10-14	50-2 50-2	50-2 50-2

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

N0.	TABLE II (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STICHING TYPE	STITCHES PER INCH	NEEDLE	THREAD BOBBIN/ LOOPER/
11.	Finish sweatband. Using the earlap-to-lining joining seam as a guide, place lower (folded edge of sweatband on top of earlap). Turn in the raw edge of sweatband on top of earlap. Turn in raw edge at front, and on outside stitch sweatband to cap through all plies 1/16 to 1/8 inch from edge (from gusset seam to gusset seam), securely tacking each end of sweatband. Do not distort the fabric on the inside of the cap.	301	Lse-1	10-14	50-2	50-2
12.	Attach forehead flap. a. Center forehead flap (loop side up) on outside front of crown, so that the overedged straight edge of flap is 1/4 to 3/8 inch up from crown edge at center and each end is even with crown-to-earlap joining seam. Stitch flap to crown through all plies, 1/4 to 5/16 inch from bottom edge, with the chain ends (see operation 4.b) folded under the flap and caught in the stitching.	301		10-14	50-2	50-2
	b. Bartack each end of flap with 1/2 inch horizontal bartacks which shall superimpose the ends of the stitching of operation 12.a.	Bartack		28 per Bartack	50-2	50-2
13.	Make earlap buttonholes. Make buttonhole in each earlap with eyelet end toward the front, positioned parallel to the sweatband stitching (canting ± 3/16 inch) with eyelet 2 inches ± 1/8 inch from the face opening and 1-5/8 inches ± 1/4 inch from the sweatband stitching (over gusset) at marks, as indicated on pattern; turn in each raw end 3/16 to 5/16 inch.	Buttonhole		76-80 per buttonhole including tack	50-2 or B	50-2 or B

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

N0.	TABLE II (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STICHING TYPE	STITCHES PER INCH	NEEDLE	THREAD BOBBIN/ LOOPER/
14	Attach back elastic. Center elastic webbing on back of earlap inch and box stitch. The box pattern shall measure 13/16 to 7/8 inch in length by 3/16 to 1/4 inch in width. Ends of stitching shall be overlapped with not less than 3 stitches. This operation may be performed on an automatic stitching machine.	301		10-14	50-2	50-2
15.	Attach fastener tapes to earlaps. Position the loop portions of the fastener tape on inside of left earlap and hook portions on the outside of right earlap as indicated by marks on pattern. Stitch each piece to the earlap along all edges (including each abutted edge) with the stitching not less than 1/16 inch from the edge. Continue stitching through the center (1/8 inch tolerance of center) with stitching parallel to the sides (see Figure 1). The end(s) of the stitching shall be overlapped not less than 1/2 inch. One side of the fastener tape shall be tangent to the binding on face opening. The line of stitching through the center of the hook tapes may be continuous by pivoting one inch from the bottom of the diagonally position tape.	301	LSbj-1	10-14	B	B
16.	Clean cap. a. Trim all thread ends to 1/4 inch maximum and remove loose threads on inside and outside of cap. b. Remove all shade markings or tickets. c. Clean all spots and stain.					

3.9.1 Finished measurements, The finished caps shall conform to measurements for each respective size, as shown in Table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

TABLE III. Finished measurements

Size	6-1/2	6-3/4	7	7-1/4	7-1/2	7-3/4	7-7/8	8	Toler-ance
Inside 1/ Circumference	21	21-3/4	22-1/2	23-1/4	24	24-3/4	25-1/8	25-1/2	+/- 1/2
Gusset width 2/	2	2	2	2	2	2	2	2	+/- 1/4

1/ Measure the inside circumference at center of sweatband without stretching.

2/ Measure across the top of gusset at top crown joining seam,

3.9.2 Workmanship. The end item shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Acceptance criteria (see 4.4).

4.2 First article inspection. When a first article is required (see 3.1 and 6.2), it will be examined for the defects specified in 4.4.3 and 4.4.4.

4.3 Acceptance criteria. Acceptance criteria will be performed as specified in the contract or purchase order.

4.3. Component and material inspection. In accordance with 4.1, components and materials will be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4 End item visual examination. The end items will be examined for the defects listed in Table IV. The lot size will be expressed in units of caps. The sample unit will be one cap.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

TABLE IV. End item visual defects

Classification				
		Major	Minor	
Examine	Defect		A	B
Material defects and damages	Any hole, cut, tear, or needle chew on outside:			
	-up to 1/8 inch in diameter or length			301
	-more than 1/8 inch but not more than 1/4 inch		201	
	- more than 1/4 inch	101		
	Any hole, cut, tear, or needle chew on inside (except required holes around buttonholes):			
	-up to 1/4 inch in diameter or length			302
	- more than 1/4 inch but not more than 1/2 inch		202	
	-more than 1/2 inch	102		
	Any mend, darn, patch, or splice	103		
	Friction burn:			
	-when fibers are broken use “hole, cut, tear, or needle chew” classification			
	-more than. 1/4 inch in length			303
	Drill hole exposed on outside			304
	Weakening defects:			
	Smash	104		
	Multiple float or loose slub:			
	-on outside		203	
	-on inside			305
	Unsightly slub or shade bar:			
	-on inside			306
	-on outside		204	
Shaded parts	Outside part shaded			307
	Forehead flap lining and earlap linings shaded			308
Cleanness	Any spot or stain on outside			309
	One or-more shade tickets not removed			310
	Any shade marking visible on outside			311
	Thread ends not trimmed, or loose threads not removed throughout cap			312
Component and assembly	Any component part or required operation omitted (unless otherwise classified herein)	105		
	Any part or required operation not as specified		205	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

TABLE IV. End item visual defects (cont'd)

Classification				
		Major	Minor	
Examine	Defect		A	B
Cutting	Any component part not cut in conformance with directional lines indicated on patterns or not in accordance with specification requirements	106		
Seams and stitching	Accuracy of seaming:			
	-seam twisted, puckered, or pleated		206	
	-part of material caught in unrelated operation or stitching	107		
	-stitching overlapping end(s) of thread break less than 1/2 inch			313
	-ends of stitching not secured as specified		207	
	Gage of stitching:			
	-irregular, i.e., unevenly gaged or corresponding stitching not uniform (to be scored only when condition exists: along major portion of seam)		208	
	-not within -range of gage specified or varies more than 1/16 inch when no range is specified			314
	Stitches broken or missing (all stitching except buttonholes).			
	-up to 1/4 inch inclusive			315
	-more than 1/4 inch-but not more than 1/2 inch		209	
	-more than 1/2 inch	108		
	Skipped stitches:			
	-1/4 inch to 1/2 inch long		210	
	-more than 1/2 inch long	109		
	Run-offs:			
	-1/4 inch to 1/2 inch long			316
	-more than 1/2 inch long		211	
	Raw edge more than 1/8 inch long			317
	Seam or stitch type not-as specified			318
	Stitches per inch (except on elastic webbing) more than maximum or less than minimum specified			319
	Stitches per inch on elastic webbing:			
	-more than maximum specified	110		
	-less than minimum specified			320
	Stitch tension (except on bartacks):			
	-loose tension resulting in a loose seam	111		

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

TABLE IV. End item visual defects (cont'd)

Classification				
		Major	Minor	
Examine	Defect		A	B
Seams and stitching (cont'd)	Loose tension (exposed loops of top or bottom threads):			
	-on outside		212	
	-on inside			321
	Tight tension (stitches break when strain is applied in the direction of the seam or stitching).		213	
	Loose tension or bartacks			322
	Overedge stitching exposed beyond finished edges of binding:			
	-on. inside			323
	-on outside	112		
Labels	Missing, incorrect, or illegible:			
	-combination identification-size label	113		
	-instruction label		214	
	Either label not stitched on all four sides			324
	Any label not positioned as specified			325
	Stitching through the printing on any label			326
	Ends of stitching overlapping less than three stitches			327
Bar code label/tag	Bar code omitted or not readable by scanner		215	
	Human readable interpretation (HRI) omitted or illegible		216	
	Not visible on folded, packaged item		217	
	Causes damage to the item	114		
Outside crown	Fleece side of forehead flap on the underside, when flap is in the up position	115		
	Straight edge of forehead flap less than 1/4 inch or more than 3/8 inch from center front edge of cap		218	
	even with crown-to-earlap joining seam by more than 1/8 inch			328
	One or both bartacks at ends of forehead flap omitted		219	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

TABLE IV. End item visual defects (cont'd)

Classification				
		Major	Minor	
Examine	Defect		A	B
Outside crown (cont'd)	Bartack(s) at ends of forehead flap less than 3/8 or more than 1/2 inch long			329
	Bartack(s) not superimposed on ends of stitching joining forehead flap to cap through all plies	116	220	
	Forehead flap not centered over face opening by more than 1/4 inch		221	
	Chain ends of overedge stitching on forehead flap not turned under flap and caught in the stitching		222	
Earlaps	Buttonhole stitching(s) not securely caught in the fabric causing stitches to pull away from fabric	117		
	Buttonhole purling(s) not on outside or ends not securely tacked		223	
	Buttonhole(s) not specified type or uncut		224	
	Buttonhole(s) omitted	118		
	One or more broken stitches or more than two skipped stitches in buttonhole		225	
	Inside edge of buttonhole eyelet(s) less than 1-7/8 or more than 2-1/8 inches from bound edge of face opening		226	
	Inside edge of buttonhole eyelet(s) less than 1-3/8 inches or more than 1-7/8 inches from sweatband stitching		227	
	Ragged- edges on buttonhole(s)			330
	Gimp omitted in buttonhole(s)		228	
	Cut length of' buttonhole(s)			
	-less than 1 inch long	119		
	-more than 1-1/8 inches long			331
	Binding twisted or stretched (wavy)			332
	Binding not securely caught in fabric (binding can be pulled from fabric)	120		
	Binding less than 1/4 or more than 3/8 inch wide			333
	Row of stitching through the center of' the hook or loop fastener tape(s) omitted	121		
	End(s) of stitching on fastener tape(s) overlapped less than 1 inch		229	
	Fastener tape stitching(s) not across each piece at abutted ends	122		
	Elastic webbing misplaced or crooked			334
	Ends of elastic webbing not box stitched		230	

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

TABLE IV. End item visual defects (cont'd)

Classification				
		Major	Minor	
Examine	Defect		A	B
Earlaps (cont'd)	Elastic webbing stitched to cap without stretching, or gusset material stretched		231	
	Elastic webbing less than 2 inches or Rounded end of hook tape more than more than 2-1/4 inches long			335
	Loop portion of fastener tape not on inside of left earlap	123		
	Hook portion of fastener tape not on outside of right earlap	124		
	Rounded end of hook tape more than 1/8 inch from binding or the abutted end(s) more than 1/8 Inch apart		232	
	Side of fastener tape(s) not tangent to binding at face opening by more than 1/8 inch			336
	Ear perforations missing		233	
	Edges or hard edges on ear perforations	125		
	Outside ply(s) tight or full	126		
Inside cap	Napped surface of crown lining faces out		234	
	Lining(s) (crown and earlap) tight or full		235	
	Sweatband twisted or puckered	127		
	Sweatband less than 7/8 inch or more			337
	Sweatband extending below crown at front of cap by 1/16 inch or more			338
	More than one seam splice in the sweatband piece		236	

4.4.2 End item dimensional examination. The end items will be examined for conformance to the dimensions specified in Table III. Any dimension not within the specified tolerance will be classified as a defect. The lot size will be expressed in units of caps. The sample unit will be one cap

4.4.3 Packaging examination. The fully packaged end items will be examined for the defects listed below. The lot size will be expressed in units of shipping containers. The sample unit will be one shipping container fully packaged.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of' application
Materials	Any component missing, damaged, or not as specified

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number of bundles per container is more or less than required Number of' caps per bundle is more or less than specified Size shown on one or more caps not as specified on outer container 1/

1/ For this defect, two bundles from each container in the sample will be examined.

4.4.4 Palletization examination. The fully packaged and palletized end items will be examined for the defects listed below. The lot size will be expressed in units of palletized unit loads. The sample unit will be one palletized unit load, fully packaged.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's or Defense System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cap is intended for wear by military personnel of the Department of Defense. It is worn alone or as a liner for the helmet assembly.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

b. Class and size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When first article is required (see 3.1, 4.3, and 6.3).
e. Packaging (see 5.1).

6.3 First article. When a first article is required, it will be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to standard samples, address the contracting activity issuing the invitation for bids.

6.4 International standardization agreements. Certain provisions of this specification are the subject of international standardization agreement as cited in NATO, STANAG No. 2333. When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

6.6 Subject term (key word) listing.

Equipment, cold climate
Hat, Arctic
Headwear

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes, with respect to the previous issue due to the extensiveness of the changes.

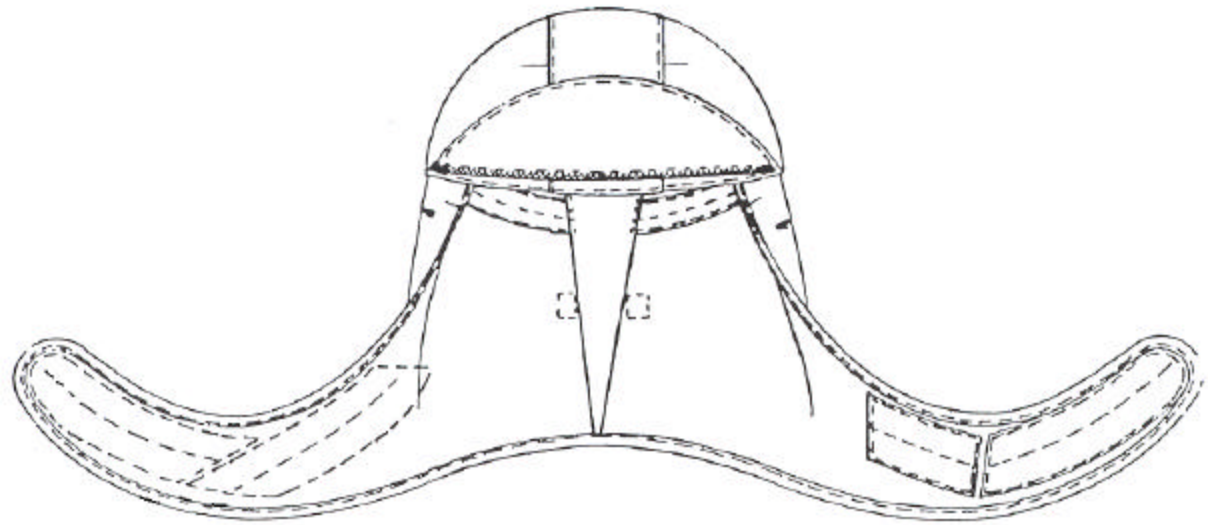
Custodians:
Army - GL
Navy - NU
Air Force - 99

Preparing activity:
DLA - CT

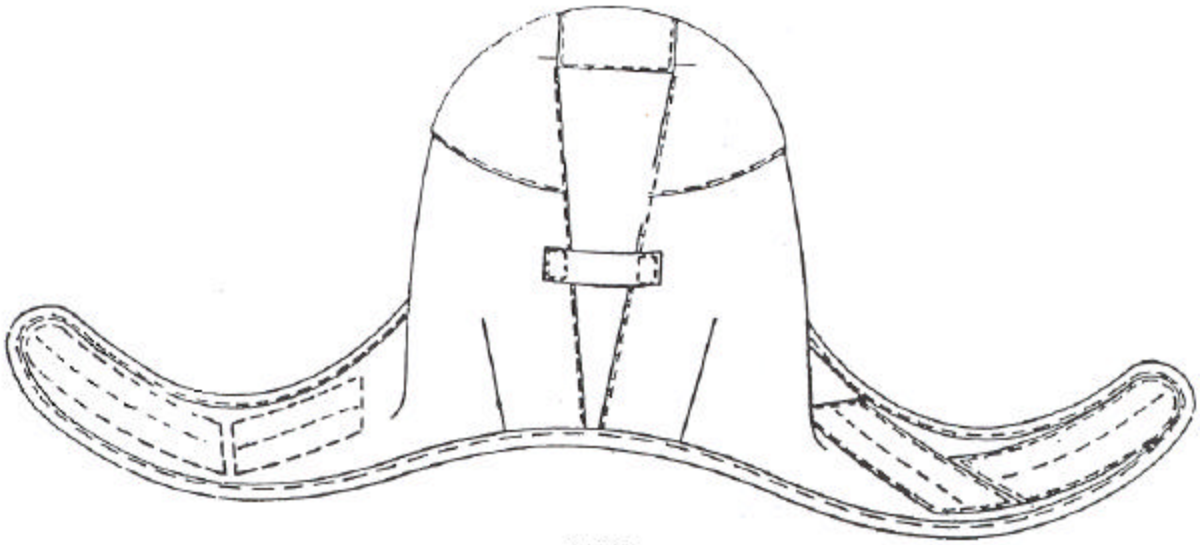
(Project 8415-0141)

Review activities:
Navy - MC
Air Force – 11.6, 82, 45

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.



FRONT



BACK

FIGURE 1. Cap, cold weather, insulating helmet liner.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-DTL-43549D	**2. DOCUMENT DATE** **2000 December 18**

DOCUMENT TITLE
CAP, COLD WEATHER, INSULATING, HELMET LINER

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON *(if applicable)*	**7.DATE SUBMITTED** (YYYYMMDD)

8. PREPARING ACTIVITY

NAME Defense Supply Center Philadelphia ATTN: DSCP-CRFD (BLDG 6, AREA D)	b. TELEPHONE *Include Area Code)* (1) Commercial 215 737 7053 (2) AUTOVON 444-7053
c. ADDRESS *(Include Zip Code)* 700 Robbins Ave Philadelphia, PA 19111- 5096	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman road, Suite 2533 Ft. Belvoir, VA 22060-2533 Telephone (703) 767-6888 AUTOVON 427-6888

DD Form 1426, FEB 1999 (EG) *PREVIOUS EDITION IS OBSOLETE*
WHS/DIOR, Feb 99

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T21:40Z
Check the source to verify that this is the current version before use.